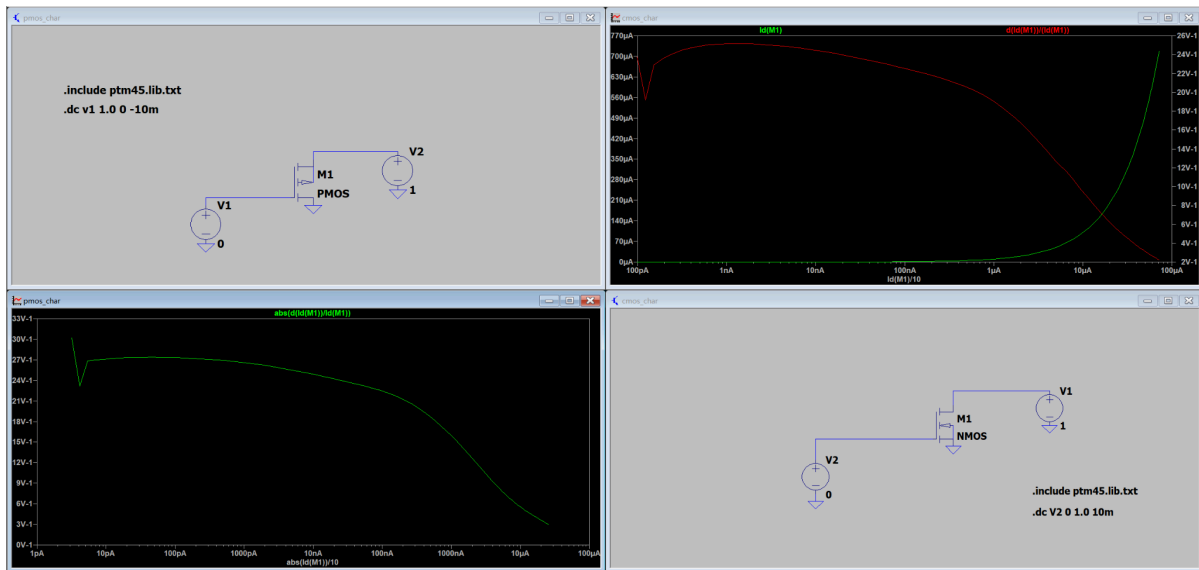
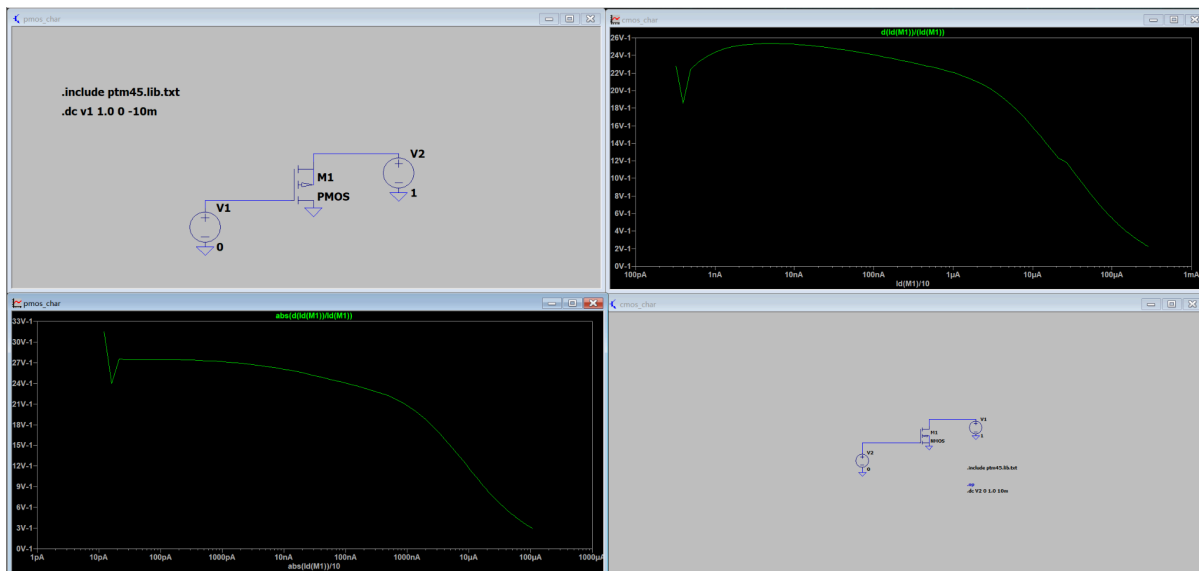


gm/ID vs. IC curve for NMOS and PMOS (W/L = 1μm/100nm)



gm/ID vs. IC curve for NMOS and PMOS (W/L = 4μm/100nm)



device characterisation for nmos (W/L = 1μm/100nm) :

```

--- BSIM4 MOSFETS ---
Name:      m1
Model:     nmos
Id:        9.95e-06
Vgs:       3.99e-01
Vds:       1.00e+00
Vbs:       2.17e-01
Vth:       4.13e-01
Vdsat:     6.41e-02
Gm:        1.87e-04
Gds:       2.14e-06
Gmb:       4.70e-05
Cbd:       4.13e-16
Cbs:       8.57e-16

```

intrinsic gain = $1.87e-4 / 2.14e-6 = 87.38$

device characterisation for nmos (W/L = 4µm/100nm) :

```

--- BSIM4 MOSFETS ---
Name:      m1
Model:     nmos
Id:        4.02e-05
Vgs:       4.04e-01
Vds:       1.00e+00
Vbs:       1.97e-01
Vth:       4.18e-01
Vdsat:     6.44e-02
Gm:        7.54e-04
Gds:       8.66e-06
Gmb:       1.87e-04
Cbd:       1.65e-15
Cbs:       3.41e-15

Total elapsed time: 0.052 seconds.
```

$\text{intrinsic gain} = 7.54\text{e-}4 / 8.66\text{e-}6 = 87.07$

device characterisation for pmos (W/L = 1µm/100nm) :

```

--- BSIM4 MOSFETS ---
Name:      m1
Model:     pmos
Id:        1.01e-05
Vgs:       4.79e-01
Vds:       1.00e+00
Vbs:       8.68e-01
Vth:       -4.55e-01
Vdsat:     -1.05e-01
Gm:        1.60e-04
Gds:       3.26e-06
Gmb:       3.56e-05
Cbd:       5.22e-16
Cbs:       6.51e-16
```

$\text{intrinsic gain} = 1.60\text{e-}4 / 3.26\text{e-}6 = 49.08$

device characterisation for pmos (W/L = 4µm/100nm) :

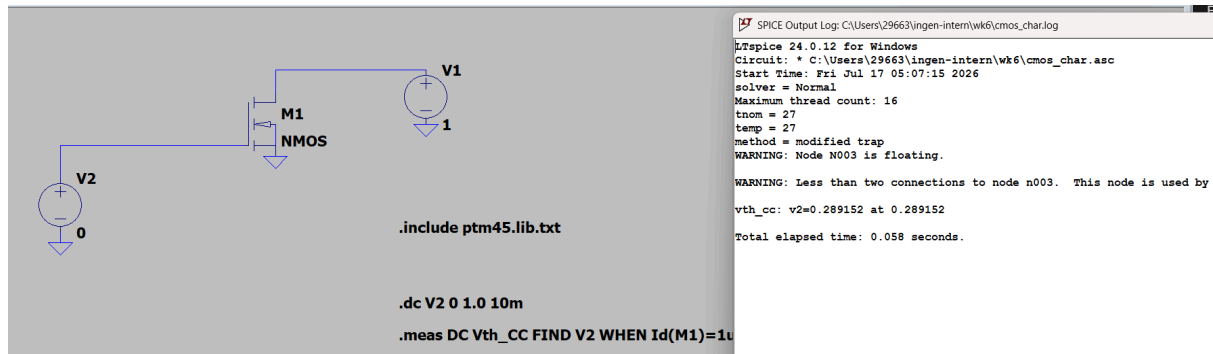
```

--- BSIM4 MOSFETS ---
Name:      m1
Model:     pmos
Id:        4.01e-05
Vgs:       4.80e-01
Vds:       1.00e+00
Vbs:       8.66e-01
Vth:       -4.55e-01
Vdsat:     -1.05e-01
Gm:        6.39e-04
Gds:       1.30e-05
Gmb:       1.43e-04
Cbd:       2.09e-15
Cbs:       2.60e-15
```

$\text{intrinsic gain} = 6.39\text{e-}4 / 1.30\text{e-}5 = 49.15$

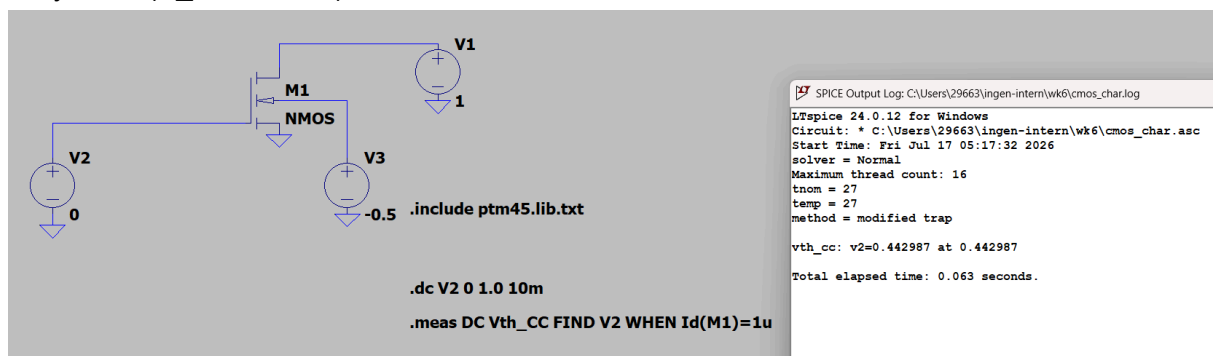
VTHextraction

nmos(W/L = 1 μ m/100nm):
constant-current method



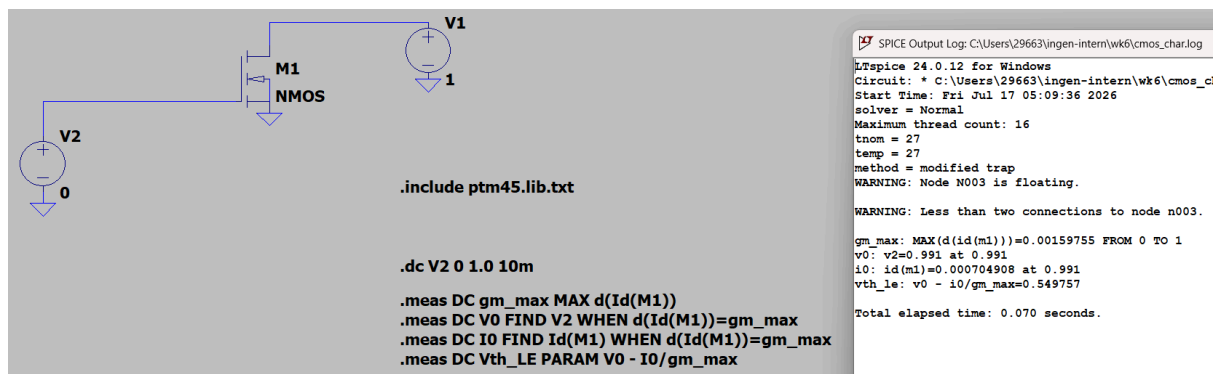
Vth = 0.29 V

body effect(V_{BS} = -0.5V):



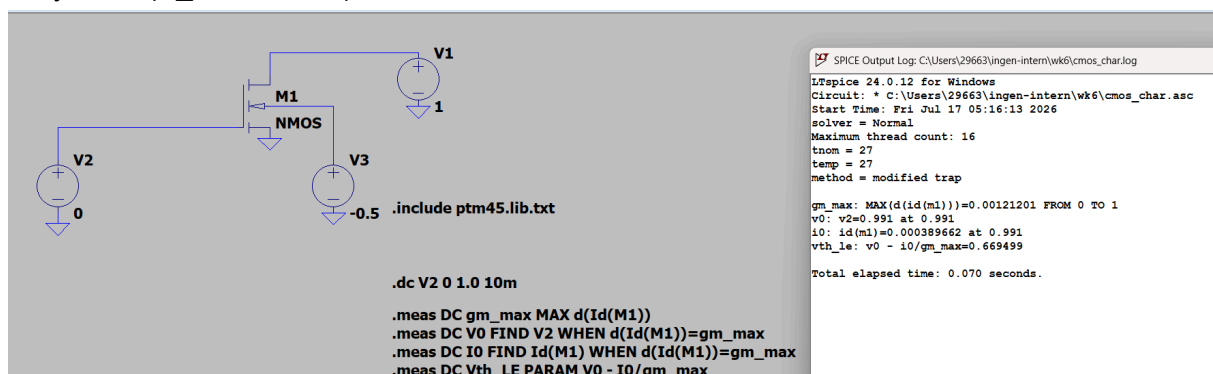
Vth = 0.44 V

linear extrapolation method



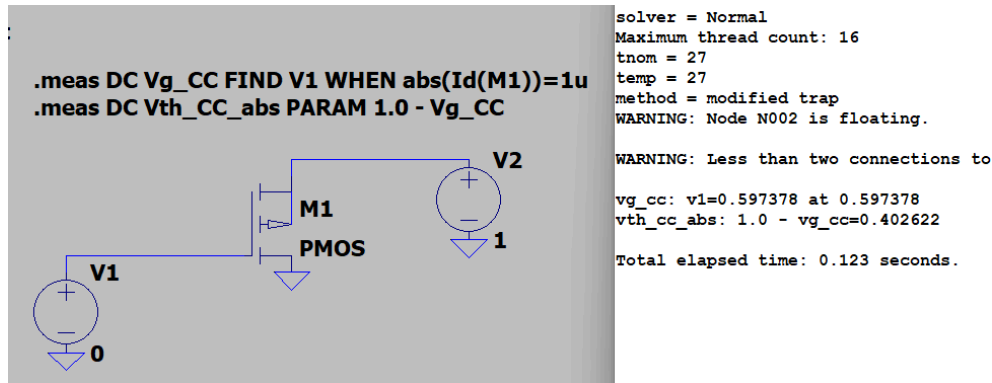
Vth = 0.55 V

body effect(V_{BS} = -0.5 V):

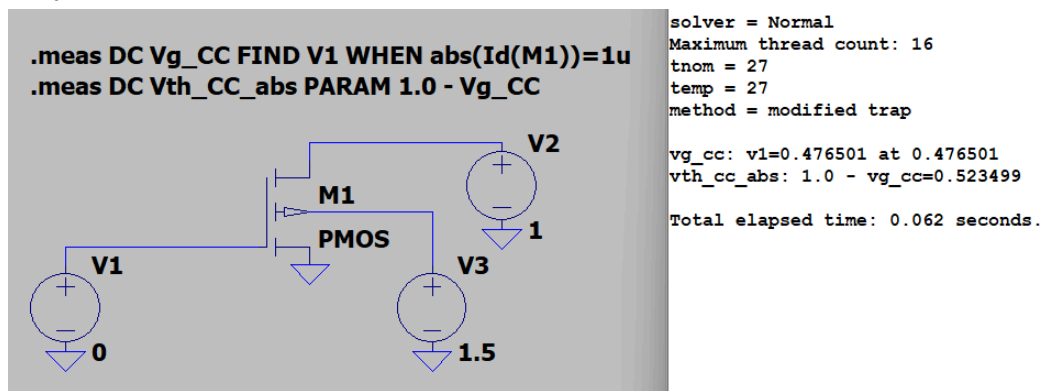


Vth = 0.67 V

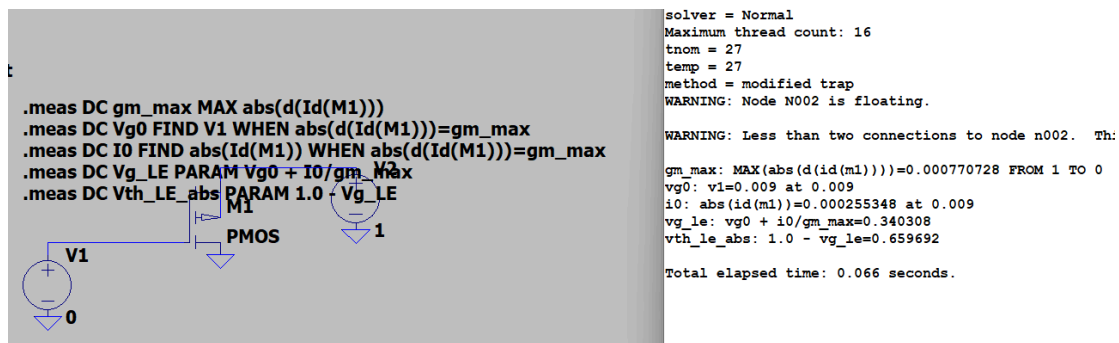
pmos(W/L = 1 μ m/100nm):
constant-current method



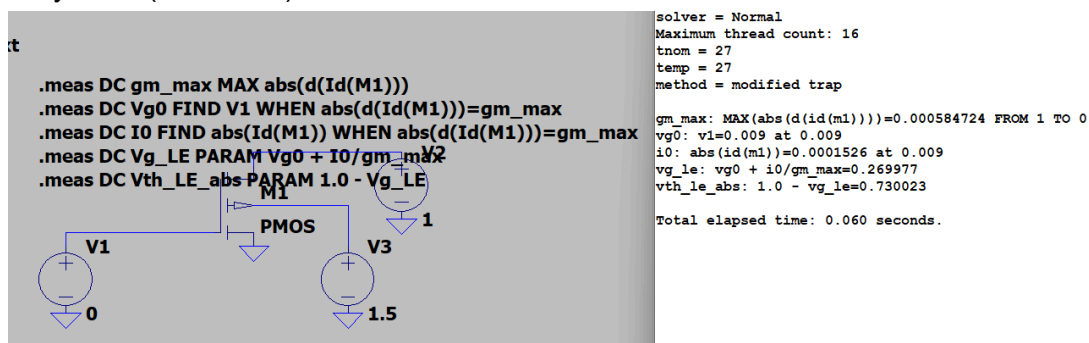
Vth = 0.4 V
body effect(Vbs=0.5V):



Vth = 0.52 V
linear extrapolation method

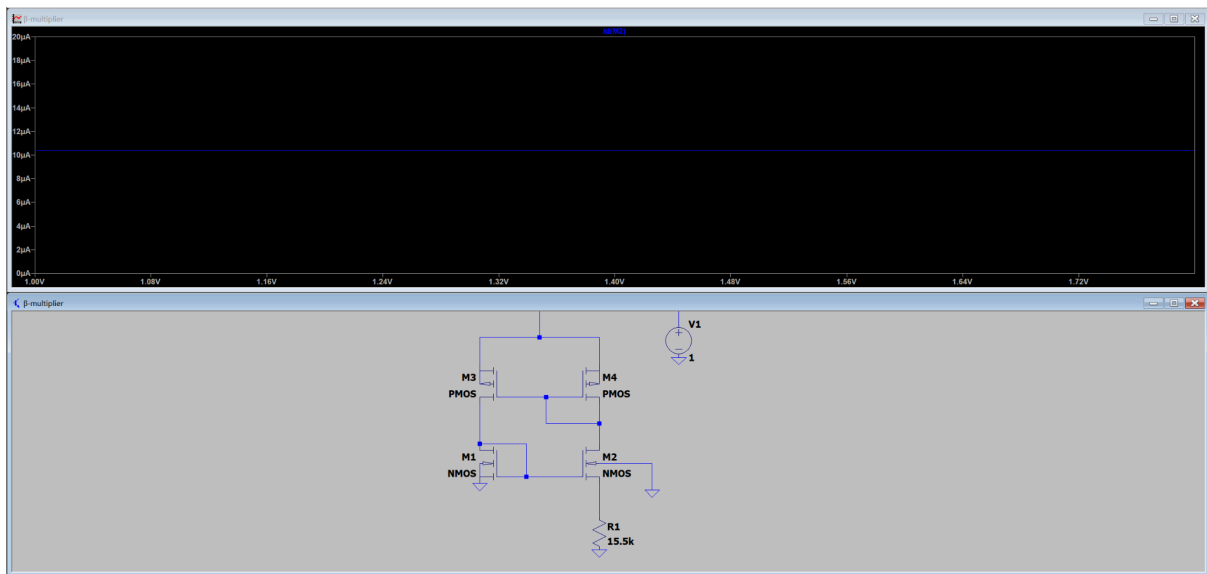


Vth = 0.66 V
body effect(Vbs=0.5V):



Vth = 0.73 V

Self-biased β -multiplier bias circuit:
Supply-independent bias :



tran startup:

